Co-Training Transformer for Remote Sensing Image Classification, Segmentation, and Detection

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***Abstract*— Several fundamental remote sensing (RS) image processing tasks, including classification, segmentation, and detection, have been set to serve for manifold applications. In the RS community, the individual tasks have been studied separately for many years. However, the specialized models were only capable of a single task. They lacked the adaptability for generalizing to the other tasks. Moreover, transformer exhibits a powerful generalization capacity because it has the property of dynamic feature weighting. Hence, there is a large potential of a uniform transformer to learn multiple tasks simultaneously, i.e., multitask learning (MTL). An MTL transformer can combine knowledge from different tasks by sharing a uniform network. In this study, a general-purpose transformer, which simultaneously processes the three tasks, is investigated for RS MTL. To build a transformer capable of the three tasks, an MTL framework named RSCoTr is proposed. The framework uses a shared encoder to extract multiscale features efficiently and three task-specific decoders to obtain different results. Moreover, a flexible training procedure named co-training is proposed. The MTL model is trained with multiple general datasets annotated for individual tasks. The co-training is as easy as training a specialized model for a single task. It can be developed into different learning strategies to meet various requirements. The proposed RSCoTr is trained jointly with various strategies on three challenging datasets of the three tasks. And the results demonstrate that the proposed MTL method achieves state-of-the-art performance in comparison with other competitive approaches. Code will be available at https://github.com/Li-Qingyun/RSCoTr.**

***Index Terms*— Co-training, multitask learning (MTL), remote sensing (RS), transformer.**

I. Introduction

With the development of imaging technology, more and more remote sensing (RS) data have been collected. In fact, RS has opened a door for Earth observing [1]. Among different types of RS, optical RS provides high-resolution images of observing surface with plentiful and complex structural information. Therefore, optical RS images are widely used in most Earth science-related disciplines including agriculture, forestry, weather prediction, land survey, and military [2].

To meet various requirements of the applications, several fundamental tasks have been set, such as scene classification, object detection, and semantic segmentation of RS images. The RS scene classification tries to allocate a pre-defined scene category to a specific RS image [3]. The RS image semantic segmentation partitions the RS image by classifying each pixel densely [4]. While for the RS image object detection, it attempts to localize and recognize the objects of interest in the RS image [5]. These fundamental RS tasks perform significant roles in real applications. Specialized models have been studied for each individual task for a long time in the RS community. However, the models were over-optimized for specific tasks and limited datasets. They lacked the generalization capacities. Besides, multiple specialized models are not parameter-efficient. Hence, in this study, we introduce multitask learning (MTL) to achieve a unified model for the fundamental tasks of RS images.

The target of MTL is to obtain a model capable of processing multiple tasks simultaneously [6]. The benefits of MTL are threefold. First, compared with task-specific paradigm, MTL models can resolve more missions with the uniform architecture and shared parameters, which is more similar to the paradigm of human perception. Second, since the data annotated for each individual task is still in short supply, the MTL expands supervised information by sharing and combining the knowledge across different tasks, which prevents the model from overfitting to data of each task as conventional method does. At last, the MTL learns joint representations of RS images which attend simultaneously to the learned tasks. The RS foundation models also attend to obtain uniform representations, which were pre-trained on a specific upstream task and fine-tuned on various tasks. Hence, the MTL will promote the development of RS foundation model studies by expanding the upstream pre-training tasks.

The designing of the MTL network architecture requires empirical referencing models for each individual task. In recent years, deep learning-based methods, especially convolutional neural network (CNN), have been widely used for various RS tasks. Due to CNN's powerful ability of feature extraction, there have been many successful CNN-based methods for each individual RS task, such as various fine-tuned CNNs for scene classification [7], UNet-like models for semantic segmentation [8], and region CNN-based methods for object detection [9]. However, CNN-based methods have been limited by the bottleneck of weak global information modeling capacity on each individual RS task.

Recently, transformer [10], which is the prevalent architecture for language processing, has been introduced for

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RS images classification [11], segmentation [12], and detection [13]. Compared with CNN, which always aggregates features in local receptive fields, transformer, which calculates attentions in the global field, is more adept at modeling the long-distance relationships in RS images. Therefore, transformer for each individual task has achieved significant promotion. Besides, the weights in transformer are calculated by attention mechanism dynamically, while the weights in CNN are stored in convolution kernel statically. Consequently, transformer has a better adaptability for learning cross-domain knowledge in MTL. But there are challenges to achieve the innovative goal with existing resources in the RS community. First, the prediction formats of the three tasks are diverse. It is invalid to design an MTL framework by easily combining arbitrary specialized transformer. It is necessary to make appropriate designing choices for the architecture empirically and combined them into a unified framework reasonably. Second, there is no training data annotated with all the three formats simultaneously. Compared to preparing a new dataset, it is more realistic to employ existing general datasets, which is commonly used to train specialized models. Third, a simple MTL training procedure of merging all single-task training is resource-intensive and uneconomical. A memory-friendly training scheme should be explored for RS MTL, which will make the MTL goal accessible to all interested researchers. Hence, in this study, we present novel and rational solutions of these critical problems.

To achieve RS MTL, a unified multitask framework based on transformer is designed. The framework has a shared RS image encoder and three task-specific decoders. Besides, a cooperatively training (co-training) procedure is proposed, which trains the MTL model with general datasets of the three RS tasks jointly. In total, we propose a method to co-training a transformer for RS tasks, namely RSCoTr. The idea is motivated by the large potential of Transformer to achieve multitask unification. In contrast to conventional task-specific approaches, the proposed method constitutes a significant advancement, delivering a threefold enhancement. First, its versatility enables support for multiple tasks and various annotation formats. Second, the model achieves more comprehensive representations of RS images by leveraging a broader range of data sources. Lastly, the approach is more economical, as it reduces the overall cost through using a shared network. To our knowledge, this study is the first to investigate MTL of RS fundamental tasks with general datasets, while other studies of MTL in the RS community mostly investigated specific MTL task with specialized datasets [14], [15].

The main contributions are summarized as follows.

1) The innovative idea of co-training a unified transformer for multiple RS fundamental tasks is introduced. Three mainstream tasks for RS images, encompassing scene classification, semantic segmentation, and object detection, are investigated in this study.
2) A novel transformer-based framework for RS MTL is proposed. The framework employs a shared encoder for efficient multiscale feature extraction and task-specific decoders to yield task-specific results.
3) A flexible co-training procedure is proposed to properly train the RS MTL model. This procedure utilizes general datasets for individual tasks, ensuring computational efficiency. Various learning strategies for co-training are explored to accommodate diverse requirements.
4) The proposed MTL method compares favorably to traditional task-specific methods, boasting versatility across tasks and annotation formats, delivering more comprehensive RS images representations, and reducing overall costs across multiple tasks.

II. Related Works

In this section, related works of this study including deep learning-based methods for RS individual tasks, transformer-based image processing methods, and MTL methods are investigated.

A. Individual Task for RS Image Classification, Segmentation, and Detection

In recent years, deep learning-based methods have become the mainstream of scene classification, semantic segmentation, and object detection of RS images. The models took advantages of mass supervision data and data-driven optimization algorithm, which achieved better generalization performance than conventional methods based on handcrafted features and unsupervised features.

In the field of RS scene classification, limited training samples were a common situation for a supervised classifier. Therefore, training a classifier from scratch might cause a serious problem called overfitting. A convenient strategy for RS image scene classification was to use an off-the-shelf model, which had been pre-trained on source domain with large-scale training dataset, such as ImageNet dataset [16]. The pre-trained CNN feature extraction network was fine-tuned on RS datasets with limited samples, which achieved higher accuracy and faster convergence [3], [7]. Moreover, Zhao et al. [17] proposed a fully pre-training strategy to prepare both a pre-trained feature extraction network and a pre-trained multilayer perceptron (MLP) classifier. Nonetheless, the data distribution diversity between nature images and RS data brought challenges to transfer learning-based methods. To address this issue, Wang et al. [18] proposed a transferring strategy based on meta-learning. Their T-CNN learned to estimate the significance of knowledge and transfer adaptively. The transfer learning significantly mitigated the overfitting problem caused by insufficient training samples in RS scene classification. Recently, the RS image classification is developing in the direction of combining multisource data. Advanced techniques have been used in multisource RS image classification studies, such as depthwise feature interaction network used in DFINet [19], and structural optimization transmission used in SOTNet [20]. These techniques also have potential for RS scene classification.

For RS image semantic segmentation, the encoder–decoder UNet with two symmetric paths was considered as a standard structure [21]. The contracting encoder gradually downsampled the input maps spatially to extract hierarchical features,

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and the expanding decoder upsampled the input maps progressively. Moreover, Diakogiannis et al. [8] designed a novel network named ResUNet, which combined the UNet with residual networks, dilated convolutions, and pyramid scene parsing networks. Li et al. [22] refactored the skip connections of the residual UNet with linear attention mechanism for fine-resolution RS image segmentation. Also based on attention mechanism, Ding et al. [23] calculated attentions of local patches on both low-level and high-level features, and embedded the attention from high-level features into low-level features to enhance semantic discrimination.

For RS image object detection, the large diversities in image spatial resolutions, target sizes, and orientations were the main challenges. To enhance the performance on various sizes of instances, especially small objects, the researchers usually conducted object detection on multiscale features. For example, Deng et al. [24] designed a multiscale region-based CNN detector, in which various sizes of region proposals were generated on multiscale feature maps and the features of regions were pooled from the fused feature. To resolve the problem of arbitrary angles, Cheng et al. [9] inserted a rotation-invariant layer to minimize the rotation variation by adding a regularization constraint. Additionally, some studies attempted to formulate object position as oriented bounding boxes, which included additional orientation information [25]. In this characteristic scope, Yang et al. [26] added an intersection over union as a constant factor to the smooth L1 loss and designed instance-level denoising, which enhanced the performance on small objects and cluttered targets. The subsequent studies are carried out from the perspective of network structure [27], angle coding [28], loss function [29], and so on. Recently, H2RBox was proposed to train a oriented detector using general horizontal annotations with weakly supervised learning [30]. Yu et al. [31] further incorporated symmetry learning to boost the supervision.

In general, the deep learning-based methods have gained huge success when performing a single RS task. The model has been dominated by CNN for a long time. However, the CNN-based methods have suffered from limited receptive field, the lack of global information and inductive bias. Transformer, which was a relatively new deep model, has shown its potential for RS image processing tasks.

B. Transformer-Based Methods

Recently, as a hot topic in the domain of computer vision, transformer was receiving more and more attention, which demonstrated its great potential in global information modeling [10]. In a variety of benchmarks, transformer-based models perform similar to or better than other types of networks such as convolutional and recurrent networks [32].

Thanks to the strong representation abilities of self-attention computations, vision transformer has been developed for RS tasks, including classification, segmentation, and detection. In the RS community, the existing methods could be roughly divided into two branches.

The methods of the first branch combined the CNN with transformer in an end-to-end manner. The local information extracted by the CNN and the long-range contextual information captured by the transformer were both obtained in a model to enhance the classification performance [11], [33], [34], [35]. For RS semantic segmentation, CNN focused on extracting local features and transformer captured consistent global information, like [36] and [37]. For RS image object detection task, Li et al. [13] designed a transformer-based framework, which adopted a CNN backbone for local feature extraction and a transformer encoder and decoder to aggregate multiscale feature maps and model the target information.

Methods of another branch combined other techniques with transformer for various RS tasks. For RS scene classification, transfer learning [38] and contrastive learning [39] were combined with transformer to enhance the classification accuracy. For RS semantic segmentation, the researchers combined U-Net with transformer as decoder, such as [12] and [40]. In addition, several works investigated swin transformer [41] for RS image object detection and instance segmentation [42].

The aforementioned transformer-based approaches focused on individual RS tasks. Their ultimate models were relatively incompetent for real applications, and they were influenced by limited supervisory information. However, learning multiple tasks within a unified transformer at the same time was still undeveloped in the RS community.

C. Multitask Learning

MTL is a machine learning paradigm inspired by human brain which can learn to handle multiple-related task simultaneously. Human can generalize the knowledge learned in a task to help learning other tasks. MTL aims to learn multiple tasks jointly so that the knowledge learned from a task can be leveraged by other tasks, which improves the generalization capability of the model [6].

Over the past decades, the MTL has achieved remarkable success in the deep learning community. Many multipurpose networks were designed to learn shared representations from multitask supervisory information [43]. The architectures mostly contained encoding components for feature extraction, and decoding components for corresponding to the annotations of each task. There were usually shared components for knowledge sharing and task interaction. According to the designing of shared components, the deep MTL networks were usually divided into two kinds: hard parameter sharing and soft parameter sharing [44].

For soft parameter-based methods, each task had a specific model with its own parameters, and the model was then regularized by other models [45]. For example, Misra et al. [46] designed a cross-stitch unit, which combined the outputs of multiple networks to learn an optimal combination of shared and task-specific representations. Ma et al. [47] proposed a multigate mixture of experts, which included several expert networks for extracting features from multiple views, and a gating network for each task to gain weight of each expert network.

The hard parameter sharing-based methods shared the parameters in hidden layers between all tasks directly. For example, Hu and Singh [48] proposed a unified transformer for vision and language tasks, whose architecture was composed

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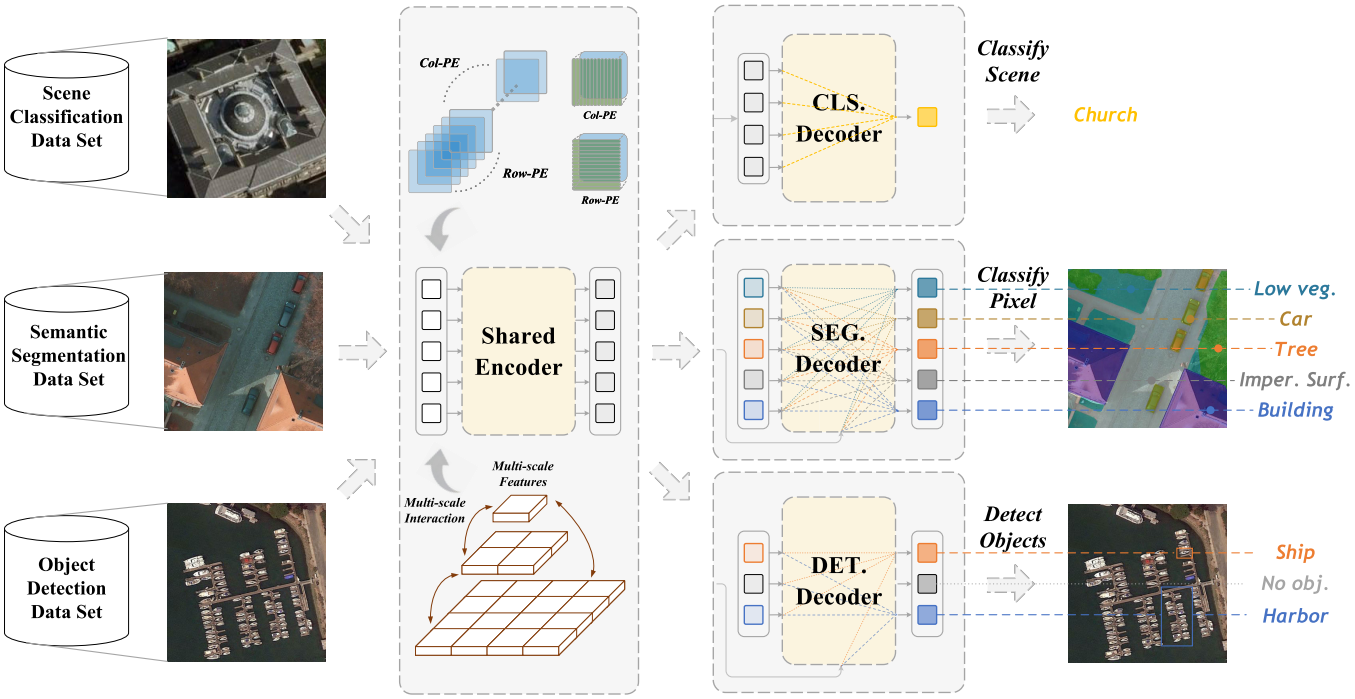


Fig. 1. Overall framework of the propose RSCoTr, which is composed of a shared encoder and three task-specific decoders. The classification decoder maps the features into a scene query. The segmentation decoder has a set of semantic queries and each query corresponds to a semantic category. The detection decoder has a set of object queries and each query corresponds to an object.

of two encoders for image and text, respectively, a hard parameter shared decoder and task-specific heads. Likhosherstov et al. [49] presented a transformer capable of multiple classification tasks of different modalities, which was composed of modality-specific tokenizers, a shared encoder and task-specific classifiers. Due to the fact that hard parameter sharing-based methods can greatly reduce the risk of overfitting, they are commonly used methods for MTL [50].

RS has also benefited from MTL. For examples, Carvalho et al. [14] proposed a CNN-based framework for learning semantics and height together, which is composed of a shared feature extractor and two independent output heads for each task. The MTL gains superior performance compared with each single task. Lu et al. [15] performed both height estimation and semantic segmentation tasks simultaneously. In this work, transformer was used for modeling global relationship and brought better performance. However, their studies were limited in the specific scopes of tasks. Bastani et al. [51] released a large-scale dataset which had seven annotation modalities and RS data with different timestamps.

In general, previous researches demonstrated the potential of MTL to achieve better performance for all tasks. Hence, it is reasonable to study MTL of RS tasks.

III. Method

In this section, the architecture details and training procedure of the proposed RSCoTr are introduced. The overall framework of the RS MTL model is shown in Fig. 1. The proposed MTL network is composed of a shared RS image transformer encoder and three task-specific transformer decoders. The shared encoder first extracts multiscale features from the input RS image. Then the features, respectively, are decoded to obtain results of the corresponding tasks.

Transformer-based models learn to accomplish a specific task with queries. A query is a feature vector that is corresponding to one prediction item. The outputs of the three tasks vary from image-level, pixel-level, and object-level predictions. To realize MTL in a unified transformer, we design task-specific queries. The classification decoder extracts a scene query from global features. The segmentation decoder has a set of semantic queries and each query corresponds to a semantic category. Besides, the object detection has a set of object queries and each query corresponds to an object.

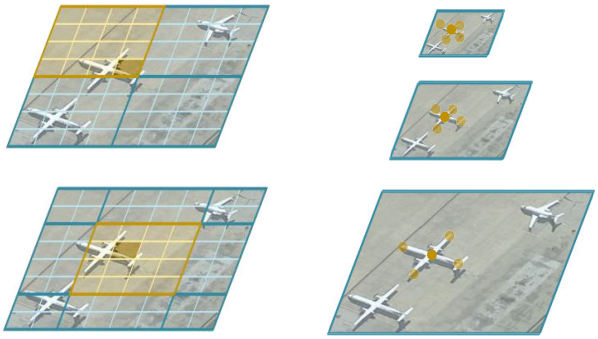
The rest of this section is arranged as follows: the shared encoder for multiscale features of RS images is provided first. Then, the task-specific decoders for classification, segmentation, and detection are described in order. At last, the well-designed co-training procedure of the RSCoTr is explained.

A. Shared Encoder of RS Image

Multiscale feature maps play essential roles in RS MTL framework. Empirically, the classification models prefer global features as summary of the image. The segmentation models require high-resolution features to parse for each pixel. The detection models need multiscale features concurrently to predict for object of various sizes. Hence, the RSCoTr requires an efficient encoder to extract multiscale features.

The multiscale feature operations bring huge computing quantity. Hence, we adopt the swin transformer [41] to obtain pyramidal feature maps and design a deformable feature fusion

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(a) (b)

Fig. 2. Two types of attention mechanisms used in the shared encoder of RSCoTr. (a) W-MSA and SW-MSA. (b) MSDA.

module to aggregate the multiscale features, which works as the feature pyramid networks [52] in CNN-based structures.

The conventional vision transformers compute multihead self-attention (MSA) of one feature query $\boldsymbol{x}_q \in \mathbb{R}^D$ with all feature points indiscriminately as in the following equation:

$$\mathrm{MSA}(\boldsymbol{x}_q) = \sum_{h=1}^{H} \boldsymbol{W}_h \sum_{p\in\Omega} \mathcal{A}_s\left(\boldsymbol{W}_{qh}\boldsymbol{x}_q, \boldsymbol{W}_{kh}\boldsymbol{x}_p\right) \cdot \boldsymbol{W}_{vh}\boldsymbol{x}_p \quad (1)$$

where $\Omega \in (0.5, 0.5), (0.5, 1.5), \ldots, (1.5, 0.5), (1.5, 1.5), \ldots)$ is the position set of feature grid. The h indexes the H heads. The p means the sampling position and the q is the position of query feature. The $\boldsymbol{x}_p$ is the sampled point feature at position p. The D is the dimension of query embedding. $\boldsymbol{W}_{qh}, \boldsymbol{W}_{kh}, \boldsymbol{W}_{vh} \in \mathbb{R}^{(D/H)\times D}$, and $\boldsymbol{W}_h \in \mathbb{R}^{D\times(D/H)}$ are learnable parameters of the hth attention head. The attention is calculated with the following:

$$\mathcal{A}_s(\boldsymbol{Q}, \boldsymbol{K}) = \mathrm{softmax}\left(\frac{\boldsymbol{K}^T\boldsymbol{Q}}{\sqrt{D/H}}\right) \quad (2)$$

where the global calculation leads to a quadratic increase of the computational complexity as the spatial resolution of the input image rises [32]. Hence, more efficient attention mechanisms are required for extracting shared features of RS images.

In the RSCoTr, two types of efficient attention mechanisms are adopted, including window multihead self-attention (W-MSA) in the swin backbone [41], and multiscale deformable attention (MSDA) used in deformable feature fusion module [53]. Fig. 2 illustrates the two types of attention mechanisms, where the position of feature query is colored dark orange, and the sampling positions of calculated locations are colored light orange.

The W-MSA and a shifted W-MSA (SW-MSA) are coupled in a swin transformer block [41]. In the W-MSA, the input feature map is split with non-overlapping windows of a fixed size. The attentions are calculated within the corresponding local window, as is formulated in the following equation:

$$\text{W-MSA}(\boldsymbol{x}_q) = \sum_{h=1}^{H} \boldsymbol{W}_h \sum_{p\in\Omega_w(q)} \mathcal{A}_s\left(\boldsymbol{W}_{qh}\boldsymbol{x}_q, \boldsymbol{W}_{kh}\boldsymbol{x}_p\right) \cdot \boldsymbol{W}_{vh}\boldsymbol{x}_p \quad (3)$$

where $\Omega_w(q)$ is the grid position set within the window for the query position q. The SW-MSA redivide features by shifting the window halfway to achieve cross-window connections, as is shown in Fig. 2(a). Hence, though they each only calculate local attentions, using both W-MSA and SW-MSA still allows global feature interaction. Besides, they have linear computation complexity with respect to the window number, i.e., the image size. They are more efficient than the conventional MSA to extract features of various resolutions.

The W-MSA and SW-MSA depend on window partition in feature map of one scale, but they cannot achieve cross-scale feature interaction. Therefore, the MSDA is used to aggregate multiscale feature maps, which is shown in Fig. 2(b). The MSDA only calculates sparse attentions at a set of essential locations around the query position $q \in \mathbb{R}^2$. Specifically, for the lth level and hth head, the sampling position set $\Omega_s(\boldsymbol{x}_q, h, l) = \{p_{hl1}, \ldots, p_{hlK}\}$ is obtained with $p_{hlk} = \boldsymbol{W}_{hlk}\boldsymbol{x}_q + q$, where $\boldsymbol{W}_{hlk} \in \mathbb{R}^{2\times D}$ learns to predict a point offset with respect to q. Then the MSDA is calculated at the sampling locations as the following equation:

$$\mathrm{MSDA}(\boldsymbol{x}_q) = \sum_{h=1}^{H} \boldsymbol{W}_h \sum_{l=1}^{L} \sum_{p\in\Omega_s(\boldsymbol{x}_q,h,l)} \mathcal{A}_d(\boldsymbol{x}_q) \cdot \boldsymbol{W}_{vh}\boldsymbol{x}_p^l \quad (4)$$

where $\mathcal{A}_d(\boldsymbol{x}_q) = \boldsymbol{W}_{qhlk}\boldsymbol{x}_q$ and $\boldsymbol{W}_{qhlk} \in \mathbb{R}^{1\times D}$. The MSDA learns to aggregate features across different levels adaptively, which is ingenious on alleviating the dramatic growth of computational complexity [53].

There are four stages in the swin backbone. Before each stage, scale and dimension transformations are performed on the feature first. Then the features are fed into the transformer blocks and maintain the dimension and quantity [41]. The outputs of all stages are fed into linear projections for a unified embedding dimension d and obtain pyramidal features $\boldsymbol{F}_l \in \mathbb{R}^{d\times H_l\times W_l}, l \in \{1, 2, 3, 4\}$, where $H_l = (H/2^{(l+1)})$ and $W_l = (W/2^{(l+1)})$. Besides, an additional global feature $\boldsymbol{F}_5$ is downsampled from $\boldsymbol{F}_4$ with a 3×3 convolutional layer. The 2-D positional encoding is embedded into the multilevel features to involve spatial information, which conducts row position encoding (Row-PE) and column position encoding (Col-PE) separately with half of the embedding dimension [13]. The MSDA then aggregates $\boldsymbol{F}_{2:5}$ and obtains the encoded multiscale features $\hat{\boldsymbol{F}}_{2:5}$. The $\boldsymbol{F}_1$ is not used for decreasing the computing quantity.

The shared encoder meets the requirements of each RS tasks. It provides features of various scales to be used for different decoder. It also performs cross-scale feature interaction to complement information of other scales. Hence, it works well for RS MTL.

B. Classification Decoder

The classification decoder is simple and easy. It obtains the scene query $\boldsymbol{q}_{\mathrm{CLS.}}$ as follows:

$$\boldsymbol{q}_{\mathrm{CLS.}} = \sum_{x=1}^{W_4}\sum_{y=1}^{H_4} \boldsymbol{F}_4(x, y), \quad \boldsymbol{q}_{\mathrm{CLS.}} \in \mathbb{R}^d \quad (5)$$

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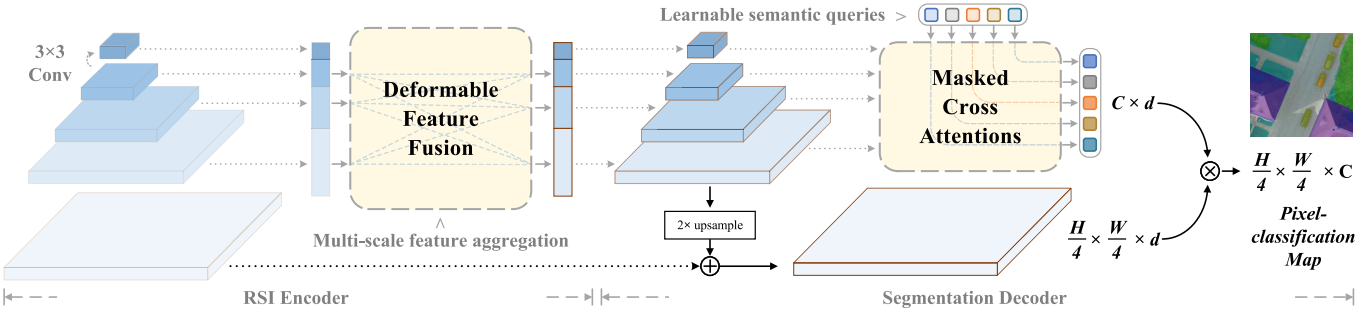


Fig. 3. Proposed architecture for segmentation task, where a high-resolution feature map is obtained from the RS image encoder, and a set of semantic queries learn to transform the feature into pixel-classification map.

then a fully connected layer with softmax operation is used to obtain the probabilities of the scene categories.

We tried several popular scheme of obtaining scene queries, such as using more linear projection, attention pooling in [54], class token learning in [32]. The performance was not better. It is empirically considered that a more complex model meets the problem of overfitting because of the limited amount and small sizes of RS images for scene classification. We also tried to explore the schemes on multiscale features $\boldsymbol{F}_{2:5}$. The performance did not promote, but the computational cost increased dramatically. Hence, we decided to perform classification with such a simplest manner.

The learning of this simple classification decoding manner is similar with the transfer learning in some studies. They first pre-trained the feature encoder with a large-scale scene classification dataset. Then they fine-tuned the encoder and a decoder of the downstream task on task-specific general datasets. However, their models suffered from the problem of catastrophic forgetting. The fine-tuned model could not perform classification, and the classification pre-training could not benefit from the other tasks. The RSCoTr co-training learns all the tasks simultaneously, which resolves the problems.

C. Segmentation Decoder

Fig. 3 illustrates the proposed architecture for the RS segmentation task. The deformable feature fusion module conducts multiscale feature aggregation on the pyramidal maps. The high-resolution feature map is intensified by adding the upsampled scale-wise feature. The segmentation decoder has a set of learnable semantic queries. Each query corresponds to a semantic category. The queries learn the pixel-level discriminant information from encoder features with masked cross-attention (MCA) modules. At last, the semantic queries transform the high-resolution feature into the predicted pixel-classification map.

The RS image segmentation task requires high-resolution dense prediction. The existing methods usually used transformed convolutions to restore the spatial resolution, or dilated convolutions to maintain the resolution [55]. The pixel-classification map was transformed from the feature map directly. However, for multitask architecture, the feature map is shared by different tasks, and should attend to pixel-level, object-level, and scene-level information simultaneously. Hence, a new scheme to obtain the pixel-classification map is deserved to be explored.

The proposed segmentation decoder learns a set of semantic queries. Each query learns a pixel-level linear transformation of a semantic category through interacting with encoder feature. In this case, the queries assume the responsibility of performing segmenting, which facilitates obtaining task-specific results without affecting the shared feature.

The structure of the proposed segmentation decoder is similar with the vanilla transformer decoder [10]. It is the cascades of identical decoder layers. Each layer contains an MSA module operating between semantic queries, an MCA module operating between encoder features and decoder queries, and a multilayer perception for query projection. Each module is wrapped with a layer normalization (LN) projection. The query embeddings are computed as the following equation:

$$\begin{aligned}\hat{\boldsymbol{X}}^l &= \mathrm{LN}\left(\mathrm{MCA}\left(\boldsymbol{X}^l\right) + \boldsymbol{X}^l\right)\\ \hat{\boldsymbol{X}}^l &= \mathrm{LN}\left(\mathrm{MSA}\left(\hat{\boldsymbol{X}}^l\right) + \hat{\boldsymbol{X}}^l\right)\\ \boldsymbol{X}^{l+1} &= \mathrm{LN}\left(\mathrm{MLP}\left(\hat{\boldsymbol{X}}^l\right) + \hat{\boldsymbol{X}}^l\right)\end{aligned} \tag{6}$$

where $\boldsymbol{X}^l$ represents the input query embeddings of the lth layer, and the $\hat{\boldsymbol{X}}^l$ represents the intermediate embeddings.

The conventional global cross-attention is considered as the source of slow convergence in recent studies [56]. Hence, the cross-attention is masked to model the context, which constrains the cross-attentions within local regions relevant to each semantic category.

Specifically, a binary attention mask $\boldsymbol{M}_{l-1}$ is predicted with the $l-1$th decoder layer. The attention map of cross-attention is masked by the former mask, as is formulated in the following equation:

$$\mathcal{A}_{\mathrm{mc}}(\boldsymbol{Q}, \boldsymbol{K}) = \mathrm{softmax}\left(\frac{\boldsymbol{K}^T\boldsymbol{Q}}{\sqrt{D_h}} + \boldsymbol{\mathcal{M}}\right) \tag{7}$$

where $\mathcal{M}$ is obtained from the mask $\boldsymbol{M}_{l-1}$ by the following equation:

$$\boldsymbol{\mathcal{M}}(x, y) = \begin{cases} 0, & \text{if } \boldsymbol{M}_{l-1}(x, y) = 1 \\ -\infty, & \text{otherwise} \end{cases} \tag{8}$$

in this manner, the semantic queries extract context features instead of attending to the whole scene.

As is illustrated in Fig. 3, when the C categories semantic queries $\boldsymbol{Q} \in \mathbb{R}^{C\times d}$ and the high-resolution feature map

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$\boldsymbol{F}_{hr} \in \mathbb{R}^{(H/4)\times(W/4)\times d}$ are obtained, they are combined into the pixel-classification map $\boldsymbol{P} \in \mathbb{R}^{(H/4)\times(W/4)\times C}$ with linear transformation, where H, W, and d indicate the high, width of the input RS image, and embedding dimension of the decoder queries, respectively. Specifically, the pyramidal feature maps $\boldsymbol{F}_l \in \mathbb{R}^{H_l\times W_l\times d}$ are inputted into the deformable feature fusion module and output fusion features $\hat{\boldsymbol{F}}_l$. The $\hat{\boldsymbol{F}}_2 \in \mathbb{R}^{(H/8)\times(W/8)\times d}$ is upsampled with binary linear interpolation. Then the $\boldsymbol{F}_{hr}$ is obtained by adding upsampled $\hat{\boldsymbol{F}}_2$ to $\boldsymbol{F}_1 \in \mathbb{R}^{(H/4)\times(W/4)\times d}$. The pixel-classification map $\boldsymbol{P}$ is calculated by the linear transformation formulated as follows:

$$\boldsymbol{P} = \boldsymbol{F}_{hr}\boldsymbol{Q}^T. \tag{9}$$

The obtained feature for RS image segmentation is both high-resolution and global-aware. The segmentation decoder employs a neoteric prediction scheme and efficient mask cross-attentions. It is competent in parsing high-resolution pixel information and making accurate prediction of pixel content for RS image segmentation.

D. Detection Decoder

The detection decoder adopts a novel query formulation as better positional priors, and an efficient denoising mission for auxiliary optimization. The detection decoder is based on the previous transformer-based RS image object detection framework [13]. The decoder had a set of object queries. Each query was used to obtain the category and location of an object. An object query was an addition of two components, a query embedding that modeled object content, and an object positional embedding that performed spatial information interaction. As is shown in the layer block of Fig. 4, the decoder was cascaded by layers composed of MSA, MSDA, and LN modules. The MSA modules learned to capture the relationship between instances. The MSDA modules learned to pool relevant object feature on multiple levels to each decoder query. In this way, each decoder layer could bring object-wise and spatial-wise perception capacities to object queries, which were fed into MLPs for objects categories recognition and boxes coordinates regression as in [57].

The object positional embeddings brought spatial priors, and were shared by all decoder layers. It was obtained from high-quality spatial-aware features. The spatial-aware features were selected by scoring encoder features with a binary classifier. The position information in the high-dimensional embeddings was implicit and uncertain, resulting in fuzzy locating and slow convergence.

It has been founded that concrete spatial prior should be provided to improve both the location capacity and training speed [58], [59]. Hence, in the proposed detection decoder, box proposals are designed as explicit prior positions. Each box $\boldsymbol{b}$ is composed of four normalized values (x_c, y_c, w, h), where x_c, y_c, w, and h are center point coordinates, width, and height, respectively. The initial boxes are predicted from the high-score encoder feature points with a fully connected network. The box proposals and their corresponding content queries are then fed into the lth decoder layers. Each output embedding predicts box offsets $\boldsymbol{\delta b} = (\delta x_c, \delta y_c, \delta w, \delta h)$ with

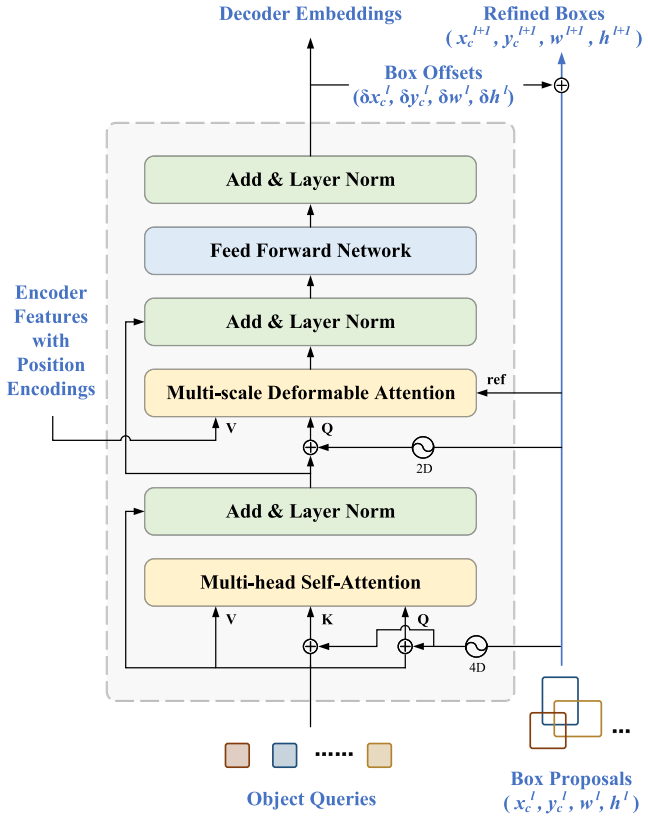


Fig. 4. Structure of each layer in the proposed detection decoder.

an MLP and calculates a new proposal for the next $l+1$th layer as the following equation:

$$\boldsymbol{b}^{l+1} = \sigma\left(\sigma^{-1}\left(\boldsymbol{b}^l\right) + \boldsymbol{\delta b}^l\right) \tag{10}$$

where $\sigma(\cdot)$ and $\sigma^{-1}(\cdot)$ indicate sigmoid and inverse sigmoid operations for transposing between normalized and non-normalized coordinations. This is to ensure that all the coordinate values are between 0 and 1. The next layer then treats the new boxes as proposals and the last output embedding as content queries. The box coordinates and features are refined gradually by the cascaded layers. Hence, the proposed detection decoder can obtain more discriminative object feature for classification and more refined boxes for location. At last, each object query from the final layer predicts certain object or background category and box coordinates.

The boxes cannot be used to calculate attentions directly, because they should be in a similar high-dimensional encoding space to the encoder position embeddings, which are represented with the 2-D positional encodings of grid points. Hence, as is illustrated in Fig. 4, for MSDA, the box centers are also encoded to compute the positional attentions between corresponding position prior and encoder feature position embeddings. Moreover, for MSA, the four coordinates are all encoded, because the positional embedding of queries and keys here are all from the 4-D encoding with a twofold embedding dimension, which does not influence attention calculation. Note that although each object positional embedding is formulated as a high-dimensional embedding, it is still explicit

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and parsable, while the original object positional embeddings cannot be decoded into definite coordinates.

The proposed detection decoder learns to refine the coarse object features and boxes iteratively, which can be simulated by the process of reconstructing noisy ground-truth labels and boxes. Hence, an auxiliary task of denoising labels and boxes is designed for training procedure. The collateral task has fixed target assigning results, so it can mitigate the effect of matching instability [60] and accelerate the convergence. Besides, to simulate predicting both positive and negative samples, both positive and negative noisy targets are generated for each ground-truth target, which provides a more reasonable optimization goal.

The noisy boxes are generated by moving the four boundaries of ground-truth boxes randomly. Specifically, the ground-truth box $\boldsymbol{b}_{\text{gt}} = (x_c, y_c, w, h)$ is converted into format $\hat{\boldsymbol{b}}_{\text{gt}} = (x_l, y_u, x_r, y_b)$, where x_l and x_r represent horizontal ordinates of left and right boundaries, y_u and y_b represent vertical ordinates of upper and bottom boundaries. The negative noisy boxes should have larger noise scale than positive ones, because the farther proposals should predict negative samples [61]. Hence, random noise offsets $\hat{\boldsymbol{\epsilon}} = (\Delta x_l, \Delta y_u, \Delta x_r, \Delta y_b)$, where $\Delta x_l, \Delta x_r \sim U(-0.5w, 0.5w)$ and $\Delta y_u, \Delta y_b \sim U(-0.5h, 0.5h)$, are generated for positive noisy boxes. While for negative ones, $\Delta x_l, \Delta x_r \sim (U(-w, -0.5w) + U(0.5w, w))$ and $\Delta y_u, \Delta y_b \sim (U(-h, -0.5h) + U(0.5h, h))$. The noisy boxes are calculated with $\hat{\boldsymbol{b}}_n = \hat{\boldsymbol{b}}_{\text{gt}} + \hat{\boldsymbol{\epsilon}}$ and then converted into format $\boldsymbol{b}_n = (x_{n_c}, y_{n_c}, w_n, h_n)$ as initial box proposals. The decoder learns to denoise $\boldsymbol{b}_n$ and reconstruct $\boldsymbol{b}_{\text{gt}}$.

The noisy labels are generated by randomly selecting part of the ground-truth labels and overlaying the selected labels with arbitrary object labels. The target labels of positive samples are assigned with the ground-truth labels, while those of negative samples are assigned with the background category.

Above all, a novel object query formulation and the denoising auxiliary task are adopted for RS image object detection. They both bring better location capacity and faster training speed to the RS image object detection.

E. Co-Training of the RSCoTr

The unified architecture of the proposed RSCoTr is capacity of performing multiple tasks simultaneously. However, in the RS community, the data which has been annotated for multiple tasks is extremely rare. Most open-source datasets are customized for an individual task. Therefore, it is essential to design a method of using the task-specific general datasets to train the RSCoTr. Additionally, there are imbalance problems among the general datasets of different tasks because they may contain images and annotations of different amounts. Moreover, the simple procedure of learning all the tasks in one training iteration requires costly computing resources. Hence, an efficient, stable, and economical training manner is required.

The designing opinions of the co-training include two aspects. On the one hand, the samples to form a batch should be annotated for the same task. Because if a batch is mixed with samples of various tasks, they have to be re-organized to go through task-specific decoders. This is equivalent to using samples of an individual task to form a batch directly. On the other hand, we only train one task in once parameter update. The reason is that training all tasks in one iteration brings huge computational cost and unpredictable optimization problems. This design of MTL makes each co-training iteration as easy as training a single task, so that no more optimization trick is required.

Algorithm 1 Proposed Co-Training Procedure

For i in $\{1, \ldots, I\}$:
  Decide training which task $t = \mathcal{T}(i)$;
  Sample a batch $(\boldsymbol{X}, \boldsymbol{Y}_{gt}) \in \mathcal{D}_t$;
  Extract shared feature $\boldsymbol{F}_{1:5}, \hat{\boldsymbol{F}}_{2:5} = \mathcal{N}_{\text{ENC.}}(\boldsymbol{X})$;
  Decode into results $\boldsymbol{Y} = \mathcal{N}_t(\boldsymbol{F}_{1:5}, \hat{\boldsymbol{F}}_{2:5})$;
  Compute loss $\mathcal{L} = \lambda_t \cdot \mathcal{L}_t(\boldsymbol{Y}, \boldsymbol{Y}_{gt})$;
  Update Parameter $\vartheta_{i+1} \leftarrow \vartheta_i - \eta \nabla_\vartheta \mathcal{L}$;
End for.

The detailed co-training procedure is shown in Algorithm 1. The datasets of the three tasks are $\mathcal{D}_{\text{CLS.}}$, $\mathcal{D}_{\text{SEG.}}$, and $\mathcal{D}_{\text{DET.}}$. The shared encoder is $\mathcal{N}_{\text{ENC.}}$ and the decoders are $\mathcal{N}_{\text{CLS.}}$, $\mathcal{N}_{\text{SEG.}}$, and $\mathcal{N}_{\text{DET.}}$. Since the traversal completions of different dataset may not occur simultaneously, the circulation unit of training procedure is designed as an update iteration, rather than an epoch. To decide the task of the ith iteration, the learning strategy $\mathcal{T} = \{t \in \{\texttt{CLS.}, \texttt{SEG.}, \texttt{DET.}\}\}$ is designed and provides the task t. The strategy is actually a pre-supposed queue of task names. Then, a batch of samples $(\boldsymbol{X}, \boldsymbol{Y}_{\text{gt}})$ is obtained from $\mathcal{D}_t$ and fed into shared encoder and corresponding decoder for prediction results $\boldsymbol{Y}$. The loss $\mathcal{L}_t(\boldsymbol{Y}, \boldsymbol{Y}_{\text{gt}})$ is calculated and modulated by a coefficient λ_t to obtain the total loss $\mathcal{L}$. At last, the model parameters ϑ_i are updated with learning rate η to obtain the new parameters ϑ_{i+1}.

The learning strategy $\mathcal{T}$ plays an essential role of controlling the co-training artificially. It is flexible to be extended into manifold strategies to meet various requirements. In this study, we propose to design the strategies by determining the learning order and frequencies of the three tasks. The order of the tasks can be constant or arbitrary. It depends on a trade-off between stabilizing training and not prioritizing a certain task; besides, the frequencies of the tasks can be identical or biased. The strategies of identical frequencies help alleviate the data imbalance problem, while the learning frequency of a more important task or a weaker task could be set higher so that the model would be more inclined to perform better on that task.

As is illustrated in Fig. 5, where each task is assigned a square with a specific color and label, we compose the two choices into four iteration strategies. For a constant order and an identical frequency, the *round robin* strategy is proposed. It trains the three tasks in turn simply. This strategy is stable and neutral. For a constant order and a biased frequency, the *repeated sequence* strategy is proposed. It trains the tasks in a pre-defined sequence and repeats it until the terminate. We can bias a certain task to meet specific requirements. For an arbitrary order and an identical frequency, the *uniform random* strategy is proposed. The randomness can avoid prioritizing a

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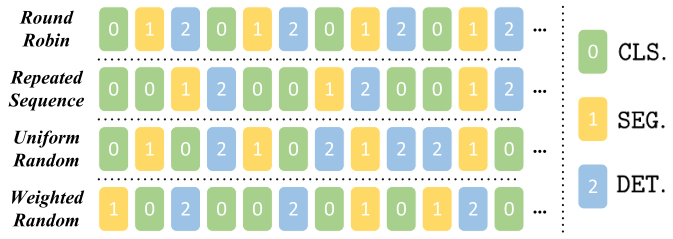


Fig. 5. Illustration of the proposed learning strategies.



Fig. 6. RS image examples of several categories in the RESISC scene classification dataset. Each fixed-sized image patch is annotated with a single label of scene.

certain task. At last, the *weighted random* strategy is presented to set probabilities of each task. It further controls the learning bias when the orders are assigned randomly. In general, the proposed co-training procedure is flexible and extensible by designing an appropriate learning strategy to meet a specific requirement.

With the well-designed co-training procedure, the RSCoTr can be trained with general datasets and acceptable computing resources. The MTL becomes convenient and efficient for RS image processing tasks.

IV. Experiment

In this section, the RSCoTr is co-trained on the three tasks with the well-designed procedure. The evaluation results of MTL are presented first. Then, the multitask learned RSCoTr is compared with single-task methods on each specific task. Finally, the shared features extracted by the proposed encoder from RS image samples of the three tasks are visualized and compared.

A. Data Description

The proposed RSCoTr is evaluated on the respective datasets of the three tasks: the RESISC [3] dataset for scene classification, the Potsdam[1] dataset for semantic segmentation, and the DIOR [5] dataset for object detection. Several training examples of the three data sets have been shown in Figs. 6–8.

1) RESISC Dataset for Scene Classification: The RESISC dataset is also called the NWPU-RESISC45 dataset. It is one of the largest datasets used for RS image scene classification researches currently. There are 31 500 annotated optical RS images of spatial resolution ranges from 30 to 2 m and a uniform size of 256 × 256. The annotations contain 45 categories, and the number of samples in each category is balanced [3]. Most of the related works chose 10% or 20% of the whole dataset randomly as the training set [18]. In this work, we adopt 20% and 80% as the ratio of the training and testing set. Besides, we randomly chose 1/8 of the testing set as validation set for aware of the training process.

2) Potsdam Dataset for Semantic Segmentation: The Potsdam dataset was released by the well-known International Society for Photogrammery and RS. The large scene of the samples covers 3.42 km^2 area of the Potsdam urban and the spatial resolution is 0.5 m. There are six categories in the annotation maps, including impervious surfaces, building, low vegetation, tree, car, and clutter. There are 24 and 14 samples in the training and testing dataset individually. The average image shape of the samples is 6000 × 6000. We preprocess the metadata as [62]. The images and the annotation maps of the original dataset are split into sample patches of 512 × 512 pixels. There are only overlaps at the end edges. The RS image in the meta dataset contains four channels of spectral bands, including near infrared, red, green, blue, and only the former three channels are adopted in our experiments.

3) DIOR Dataset for Object Detection: The DIOR dataset is one of the most challenging datasets for studying multiclasses object detection methods on optical RS images. It is large scale for 23 463 optical RS image samples, 1 92 472 annotated objects, and 20 categories of instances. Besides, there are large variations of the inter-class and intra-class object sizes and spatial resolutions. Besides, the image scenes and qualities are diverse, because of different imaging conditions and environmental climates. The images have the common size of 800 × 800 pixels and spatial resolutions of 0.5–30 m. We follow the official setting of splitting datasets. The image numbers of the training set, the validation set, and the testing set are 5862, 5863, and 11 738, respectively [5].

B. Evaluation Metrics

For evaluating the quantitative performance of the proposed RSCoTr on various tasks objectively, the evaluation results of multitask learned RSCoTr are presented and compared with task-specific methods, following metric protocols of different tasks. Specially, the overall accuracy (OA) is adopted as the metric of scene classification; The per class F_1 score, the mean F_1 score (mF_1), and OA are calculated and reported for semantic segmentation; The per class average precision (AP) and the mean AP (mAP) are utilized to assess the performance of object detection.

The OA is used for both scene classification and semantic segmentation. For the former, it is the ratio of the correctly classified image number to the overall samples amount in the test dataset, while for the later, it indicates the sum of all correctly classified pixels numbers divided by the total number of pixels in all image patches. The OA can reflect the overall performance of the scene or pixel classifier for the test dataset.

The F_1 score is defined as the harmonic mean of the precision and the recall for a specific category. For C-class classification, the calculation formulas of the mF_1 score and

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

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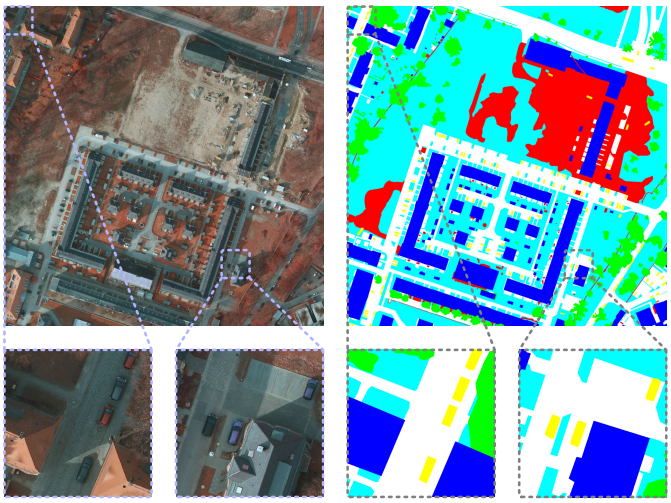


Fig. 7. Examples of a training sample and its segmentation map in the Potsdam semantic segmentation dataset and two split patches of it.



Fig. 8. Examples of RS images with ground-truth boxes in the DIOR object detection dataset. The target number and target sizes vary greatly in different RS images.

the cth class F_1 score are (11) and (12), where TP_c, FP_c, and FN_c stand for the numbers of true positive, false positive, and true negative samples in the cth class, respectively. And the P_c and the R_c stand for precision and recall values of the cth class, respectively. They weigh precision and recall with the same importance, and are comprehensive results of the two values

$$mF_1 = \frac{1}{C}\sum_{c=1}^{C} F_{1_c} \tag{11}$$

$$F_{1_c} = \left(\frac{1}{2} \times \left(\frac{1}{P_c} + \frac{1}{R_c}\right)\right)^{-1} = 2 \times \frac{P_c \times R_c}{P_c + R_c} \tag{12}$$

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}. \tag{13}$$

The calculation of per-class AP and mAP also depends on the precision and recall. A precision–recall curve $P_c(R)$ can be obtained by setting different thresholds of the box confidence to obtain a set of precision and recall values under various conditions. The AP of the cth class AP_c is the area under the P–R curve of the class $P_c(R)$, as in (14). And the mAP is the average of the AP_c of the C classes, as in the following equation:

$$AP_c = \int_0^1 P_c(R)\mathrm{d}R \tag{14}$$

$$mAP = \frac{1}{C}\sum_{c=1}^{C} AP_c. \tag{15}$$

C. Implementation Details

1) Network Settings: The swin-tiny is adopted as the shared backbone of the RSCoTr, i.e., the window size is set 7 and the patch size is set 4 [41]. The layer number is set 6 for detection decoder and nine for segmentation decoder. The embedding dimensions of the decoders are all set 256.

2) Data Preprocessing: The data is preprocessed with different transforms according to task-specific protocols during the training stage. For scene classification, a series of data augmentations used in [41] and [62] are applied to enhance the performance. For object detection, random scale and random crop augmentation as in [57] are adopted. For semantic segmentation, only random crop and random flip augmentations are used.

3) Optimization Settings: The AdamW with a momentum coefficient of 0.9, and a weight decay of 10^{-4} is adopted as the optimizer. The initial learning rate is set 5×10^{-6} for the backbone, and 5×10^{-5} for the other components. The number of the co-training iterations is set 3 00 000. The learning rate underwent twice tenfold downward adjustments throughout the training process, at 2 40 000 and 2 85 000 iterations, respectively. The batch sizes are set 16, 2, 1 for classification, segmentation, and detection, respectively.

4) Loss Functions: The well-designed training procedure allows loss functions customized for each task. In the experiments, the cross-entropy loss is used for scene classification and semantic segmentation. For object detection, the L1 loss and generalized intersection over union (GIoU) loss are used for bounding box coordinates regression and the focal loss is used for bounding boxes classification and proposal scoring [63].

5) Parameter Initialization: To accelerate the training of the RSCoTr, we empirically pre-train the entire model on detection task before co-training. Because the object detection task is the most difficult task to optimize among the three tasks. For the pre-training stage, the backbone weights are initialized with open-source weight for ImageNet dataset [16]. The learning rates are the same with the initial learning rates of the co-training, and the epoch is set 50.

6) Experimental Environment: The experiments are all conducted on scientific computing workstations with Intel Xeon Silver 4210R CPUs and Tesla V100 MAX-Q GPUs. All the training experiments are executed with single GPU on non-distributed mode.

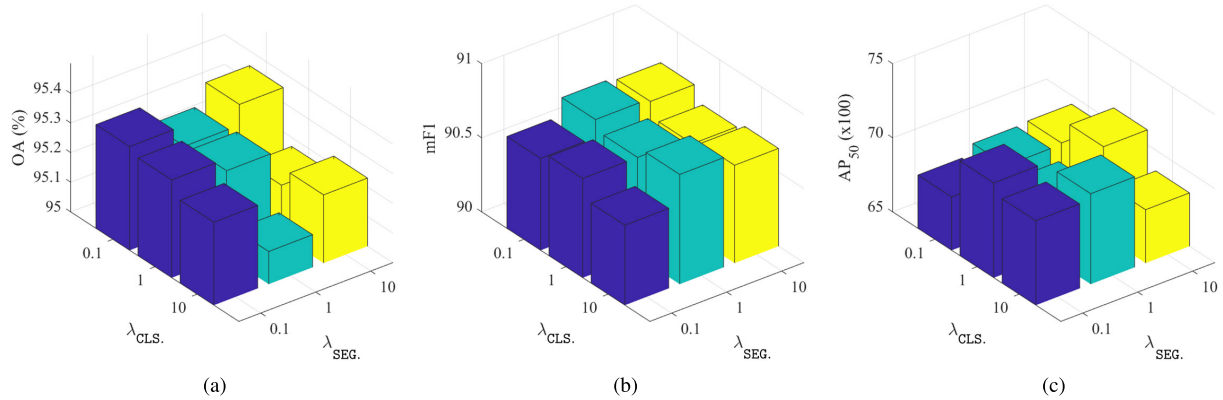
D. Results of the Proposed Multitask RSCoTr

We co-train the three tasks with the proposed RSCoTr under different settings of iteration strategies and loss coefficients. Both the quantitative and qualitative results of are reported and analyzed.

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TABLE I
COMPARISON RESULTS OF THE SINGLE TASK BASELINE AND THE PROPOSED RSCOTR TRAINED WITH DIFFERENT LEARNING STRATEGIES

Strategy	Stable order	Identical Frequency	Classification	Segmentation		Detection
			OA	OA	mF_1	mAP
Single task baseline	-	-	94.82	90.13	89.68	70.2
Batch-weighted random			92.40	90.68	90.60	69.4
Repeated sequence	✓		**95.50**	90.67	90.62	62.4
Uniform random		✓	95.34	90.67	90.64	69.1
Round robin	✓	✓	95.33	**90.70**	**90.67**	**71.4**



(a) (b) (c)

Fig. 9. Quantitative results of the proposed RSCoTr trained with different settings of loss coefficients. (a) Classification results. (b) Segmentation results. (c) Detection results.

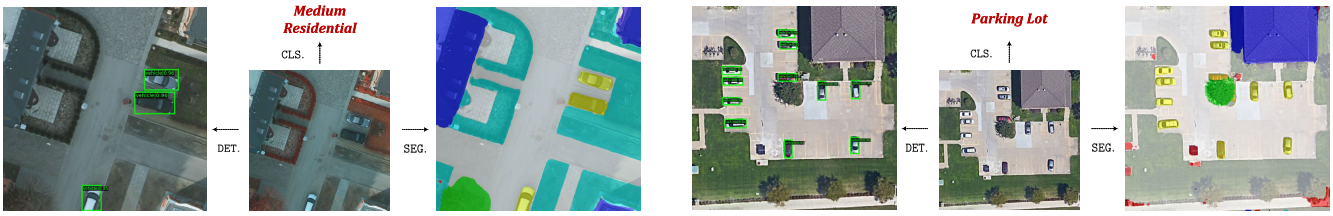
1) Various Iteration Strategies: The strategy should be designed to meet certain specific requirement and setting. In this study, we conduct empirical studies on the three fundamental RS image processing tasks. And we tend to achieve a neutral and overall better performance. Table I presents the comparison results of the proposed learning strategies and baselines. The *single-task baseline* indices only training the RSCoTr on the dataset of a single task. In this case, all the iteration milestones will be compressed to one-third in the training process. And the other settings, like initial learning rates, weight decay, and so on, remain the same with those of co-training for comparison. For *repeated sequence* strategy, we empirically consider that the classification and segmentation tasks require longer training than detection task. Therefore the sequence is set `DET.-SEG.-SEG.-CLS.-CLS.-CLS.` as an example. The *batch-weighted random* is a special case of the *weighted random* strategy, which uses the numbers of training batches as frequencies. Note that the models of different tasks in *single task baseline* are independent, while the network is shared by all tasks in other multitask models.

It can be seen that the *round robin* strategy achieves the best overall performance. The classification accuracy of the *batch-weighted random* strategy is the lowest because the batched number and frequency of classification are small. For the *repeated sequence* strategy, the classification score is the highest because the strategy is biased toward performing classification better. The consequence is that the mAP of the weakened detection task drops dramatically. The *uniform random* and *round robin* strategies both achieve decent overall performance. With an identical frequency, the round robin strategy achieves the best performance on segmentation and detection tasks and comparable performance on classification task. Consequently, the co-training strategy is better to equalize the frequency of each task and maintain a stable iteration order. Besides, with a proper strategy, the advantages of larger data amount and shared knowledge bring better performance than each individual tasks. This demonstrates the large potential of RS MTL.

To further verify the effectiveness of the proposed procedure to deal with various imbalanced data settings, the RSCoTr is trained with the round robin strategy on the half of the DIOR detection dataset and the full of the other two datasets. The other settings remain unchanged. The results are presented in Table II. The performance declines on all the three tasks because of less overall training data. But the performance on each task is still better than the each single-task baseline. The results demonstrated that the proposed co-training strategy is robust to mitigate the data imbalance problem so that to exhibit stable and effective performance.

2) Loss Coefficient of Each Task: The relationship among the three coefficients, $\lambda_{\texttt{CLS.}}$, $\lambda_{\texttt{SEG.}}$, $\lambda_{\texttt{DET.}}$, is relative. Hence, the $\lambda_{\texttt{DET.}}$ is fixed to 1, and the $\lambda_{\texttt{CLS.}}$ and $\lambda_{\texttt{SEG.}}$ are both scaled by the presupposed scale factors, including 0.1, 1, and 10. The *round robin* strategy is adopted as the learning strategy. Fig. 9 shows the quantitative results of the proposed RSCoTr under different settings of loss coefficients. The OA, mF_1, and mAP are adopted as representative metrics of the classification, segmentation, and detection tasks, respectively. First, the classification and segmentation results are more stable than the detection result, since the ratios of the minimum value

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(a) (b)

Fig. 10. Qualitative results of the multitask learned RSCoTr predicting multiple tasks for one sample. (a) Multitask results of a sample in the Potsdam segmentation dataset. (b) Multitask results of a sample in the DIOR detection dataset.

TABLE II
COMPARISON RESULTS OF THE PROPOSED RSCOTR CO-TRAINED ON DIFFERENT DATASETTINGS

Data setting	Single Detection	MTL Detection	Classification	Segmentation	
	mAP	mAP	OA	OA	mF1
Full DIOR	70.2	71.4 (+1.2)	95.33 (+0.51)	90.70 (+0.57)	90.67 (+0.99)
Half DIOR	64.8	65.4 (+0.6)	95.30 (+0.48)	90.35 (+0.22)	90.25 (+0.57)

to the maximum value of the metrics are 99.75%, 99.78%, and 96.08% for classification, segmentation, and detection, respectively. Second, the optimal coefficients set are specific for various tasks. The best classification performance appears when $\lambda_{\text{CLS.}}$ is 0.1 and $\lambda_{\text{SEG.}}$ is 0.1 or 10. The segmentation results are best when $\lambda_{\text{SEG.}}$ is 1 and $\lambda_{\text{CLS.}}$ is 0.1 or 10. And the detection achieves the best results while $\lambda_{\text{CLS.}} = 1 and \lambda_{\text{SEG.}} = 0.1$. Moreover, if we treat the average values of the representative metrics as the overall performance of the MTL, the best performance appears while $\lambda_{\text{CLS.}} = 1$, $\lambda_{\text{SEG.}} = 0.1$.

3) Qualitative Results: The multitask learned RSCoTr has the capacity of predicting results of multiple tasks simultaneously. Hence, we select two RS image samples from the datasets as the input and obtain the prediction results, which is displayed in Fig. 10. The sample in Fig. 10(a) is a patch split from a sample of the Potsdam segmentation dataset. Note that the annotation of this sample contains only segmentation maps without scene category and target location information. There are three cars, parts of residential buildings, green belts around the buildings, and branches of a tree in the picture. From the outline of the building, the width of the road and the spatial distribution, the picture may be in a residential area. The RSCoTr classifies the patch as the category "Medium Residential." The three cars are detected and segmented accurately. The model also segments for the regions of buildings and the tree. The sample in Fig. 10(b) is from the DIOR detection dataset, whose annotation does not contain the scene category and segmentation map. There are 11 vehicles parked in the indicated line. There is a tree at the image center and a building in the upper right of the picture. It can be judged from the ground pattern that the picture is a parking lot. The multitask learned RSCoTr classifies correctly. Besides, it segments the car, the tree, the building, and detects all vehicles. The samples in the RESISC classification dataset are not adopted because their spatial resolutions are low and not appropriate for dense prediction. It can be demonstrated from the qualitative results that the proposed RSCoTr can perform multiple tasks well simultaneously.

TABLE III
OVERALL ACCURACIES OF THE PROPOSED RSCOTR AND COMPARISON CLASSIFIERS ON THE RESISC CLASSIFICATION DATASET

Type	Method	OA
CNN -based	MF2Net [64] (GRSL20)	92.73
	IDCCP [65] (TGRS21)	93.76
	GRMANet [66] (TGRS21)	94.72
	DFINet-SC [19] (TGRS22)	93.10
	M2FN [67] (GRSL22)	94.50
	RSP-ResNet-50 [62] (TGRS22)	95.02
Transformer -based	DeiT-Base [11] (ICML21)	93.83
	Swin-Base [41] (ICCV21)	94.04
	IMP-Swin-T [62] (TGRS22)	94.70
	RSP-Swin-T [62] (TGRS22)	94.51
	SCViT [33] (TGRS22)	94.66
	L2RCF [68] (TGRS23)	95.24
Proposed Methods	**RSCoTr-CLS.**	94.82
	RSCoTr	**95.33**

The highest results and the names of proposed methods are bold-faced.

E. Compare With Single-Task Methods

We adopt the RSCoTr learned with *round robin* strategy when $\lambda_{\text{CLS.}} = 1$, $\lambda_{\text{DET.}} = 1$, and $\lambda_{\text{SEG.}} = 0.1$. The MTL model are compared with the methods specialized for each individual task. For scene classification, both the accuracies and the confusion matrix are presented. For semantic segmentation and object detection, the comparisons of both quantitative and qualitative results are provided.

1) Scene Classification: The proposed multitask learned RSCoTr is compared with six CNN-based models, six transformer-based models, and a baseline method. The baseline of RSCoTr architecture is trained individually on classification dataset, i.e., RSCoTr-CLS.. The DFINet-SC is adapted from DFINet based on depthwise feature interaction by replacing data source with scene classification data [19].

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TABLE IV
QUANTITATIVE RESULTS OF THE PROPOSED RSCOTR AND COMPARISON SEGMENTORS ON THE POTSDAM SEGMENTATION DATASET

Method	OA	mF_1	F_1 score per category				
			Imper. surf.	Building	Low veg.	Tree	Car
FCN [4]	89.42	88.66	91.46	**96.63**	85.99	**86.94**	82.28
DANet [69]	89.72	89.14	91.61	96.44	**86.11**	**88.04**	83.54
PSPNet [70]	89.45	**90.51**	91.61	96.30	**86.41**	**86.84**	91.38
UFMG_4 [71]	87.90	89.50	90.80	95.60	84.40	84.30	**92.40**
RSP [62]	**90.78**	90.03	**92.65**	96.35	86.02	85.39	89.75
DeepLabv3+ [23]	89.74	**90.94**	92.35	**96.77**	85.22	86.79	**93.58**
RSCoTr-SEG.	**90.33**	90.30	**92.66**	96.31	85.81	85.22	91.51
RSCoTr	**90.70**	**90.67**	**92.84**	**96.60**	**86.21**	85.87	**91.81**

The highest three results and the names of proposed methods are bold-faced.
Abbreviations: Imper. surf.: Impervious Surfaces; Low veg.: Low Vegetation.

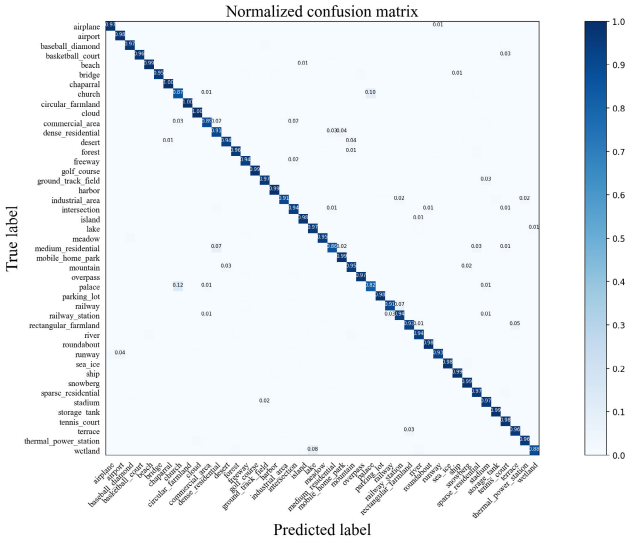


Fig. 11. Confusion matrix of the proposed RSCoTr evaluated on the RESISC classification dataset. The values have been normalized.

The L2RCF is designed to achieve better collaborations between visual transformer and CNN for scene classification [68]. The results are in Table III. As comparison, the transformer-based methods do not perform a significant advantage over the CNN-based method. Some transformer-based models with larger amount of parameters such as DeiT-Base and Swin-Base may not perform well due to the overfitting caused by insufficient training samples. By transferring the pre-trained parameters from the ImageNet and the large-scale RS scene classification dataset, the IMP-Swin-T and RSP-Swin-T achieve better results than the Swin-Base with the same design and less parameters. The RSCoTr-CLS. achieves similar result with the IMP-Swin-T, because the backbone structures are the same and there are tiny differences on prediction layers. With diverse knowledge and tasks collaboration, the RSCoTr surpasses the single-task baseline and the state-of-the-art transformer-based L2RCF by 0.51% and 0.09%, respectively. Fig. 11 shows the normalized confusion matrix of the proposed RSCoTr. The predictions are concentrated on the diagonal of the matrix and sparse on the rest parts, which imply the good classification performance.

2) Semantic Segmentation: For semantic segmentation, five RS image scene parsing models are adopted as comparison. The FCN [4], DeepLabv3+ [69], and PSPNet [70] are all influential semantic segmentation models. The DANet proposed the position attention module and the channel attention module to integrate local and global features adaptively [69]. The UFMG is an RS image segmentor that trains a dilated network to extract multicontext features from heterogeneous contexts. The RSP is the state-of-the-art RS image segmentation method which pre-trained the Swin-Tiny backbone with large-scale aerial scene dataset and fine-tuned it in an efficient UperNet framework for segmentation. To verify the effectiveness of MTL, the proposed RSCoTr is also trained on the Potsdam segmentation dataset individually as a comparison, i.e., RSCoTr-SEG.. It can be seen that the proposed methods deliver results on par with or better than state-of-the-art results. The proposed multitask learned RSCoTr achieves significantly higher mF_1 and per-class F_1 than RSP with the same backbone architecture. Additionally, the MTL brings to the segmentor 0.37% enhancement of OA and 0.37% promotion of mF_1, which demonstrates that the proposed MTL procedure takes effect of tasks interaction. From Fig. 12, the proposed methods can be seen to classify each pixels precisely.

3) Object Detection: Table V presents the quantitative comparison results on the DIOR detection dataset. Four competitive CNN-based detectors and one state-of-the-art transformer-based detector are selected as comparison. The YOLO v3 is a high-efficiency one-stage detector [72]. For comparing detection performance with multiscale features, the FPN [52] is implemented on both the Faster RCNN [73] and Mask RCNN [74] as comparisons. The T-TRD-DA is a transformer-based detector with multiscale feature fusion, transfer learning, and more efficient data argumentation [13]. The MSFGNet is a state-of-the-art CNN-based anchor-free object detection framework with multiscale features [75]. The GLNet is a novel framework which uses clip long short-term memory to enrich feature semantic and adaptive anchors to resolve object scale problem [76]. The RSCoTr-DET. is obtained by training the proposed RSCoTr architecture individually on the detection dataset. As the results shows, the methods detecting with multiscale features

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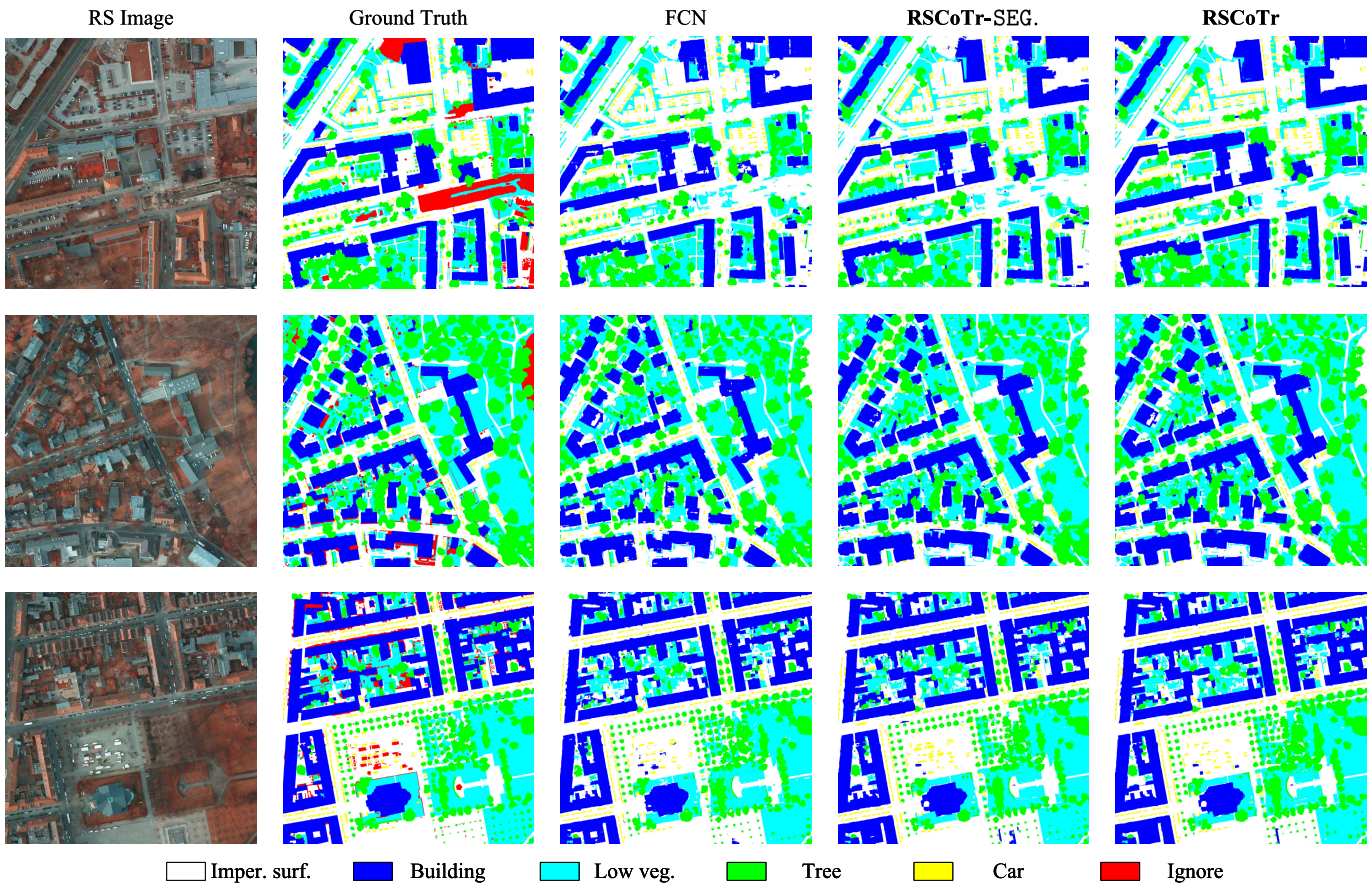


Fig. 12. Qualitative results of the multitask learned RSCoTr, the single-task learned RSCoTr-SEG., and a deep learning-based comparison method (FCN [4]). The pixels of 11Ignore' category are ignored in the evaluation, therefore, are painted with white in the segmentation results, as is implemented in [62].

TABLE V
QUANTITATIVE RESULTS OF THE PROPOSED RSCOTR AND COMPARISON DETECTORS ON THE DIOR DETECTION DATASET

	Method	YOLO v3 [72]	Faster RCNN with FPN [73]	Mask RCNN with FPN [74]	T-TRD-DA [13]	MSFGNet [75]	GLNet [76]	**RSCoTr-DET.**	**RSCoTr**
AP (×100) for Each Class	Airplane	72.2	63.2	53.8	**77.9**	**85.8**	62.9	75.8	**79.1**
	Airport	29.2	61.3	72.3	80.5	76.2	**83.2**	**83.2**	**83.7**
	Baseball field	74.0	66.3	63.2	70.1	**74.4**	72.0	**78.2**	**79.9**
	Basketball court	78.6	**85.5**	81.0	**86.3**	**90.1**	81.1	85.1	84.6
	Bridge	31.2	36.0	38.7	39.7	**44.2**	**50.5**	44.1	**46.2**
	Chimney	69.7	73.9	72.6	77.9	78.1	**79.3**	**82.5**	**82.3**
	Dam	26.9	45.0	55.9	59.3	55.5	**67.4**	**65.5**	**68.9**
	ESA	48.6	56.9	**71.6**	59.0	60.9	**86.2**	62.7	**65.7**
	ETS	54.4	49.0	**67.0**	54.4	**59.5**	**70.9**	57.3	57.8
	Golf course	31.1	73.2	73.0	74.6	76.9	**81.8**	**78.2**	**78.6**
	GTF	61.1	67.5	75.8	73.9	73.7	**83.0**	**77.1**	**78.2**
	Harbor	44.9	48.9	44.2	49.2	49.6	**51.8**	**59.5**	**58.4**
	Overpass	49.7	54.7	56.5	57.8	57.2	**62.6**	**60.3**	**60.5**
	Ship	87.4	73.2	71.9	74.2	**89.6**	72.0	**89.1**	**88.9**
	Stadium	**70.6**	62.8	58.6	61.1	69.2	**75.3**	63.9	**73.3**
	Storage tank	68.7	68.3	53.6	69.8	**76.5**	53.7	**79.5**	**78.5**
	Tennis court	**87.3**	78.7	81.1	**84.0**	**86.7**	81.3	83.6	83.3
	Train station	29.4	51.4	54.0	58.8	51.8	**65.5**	**62.8**	**66.1**
	Vehicle	48.3	48.1	43.1	50.5	**55.2**	43.4	**55.9**	**53.9**
	Wind mill	78.7	70.1	**81.1**	77.2	**84.3**	**89.2**	79.0	76.9
mAP (×100)		57.1	61.7	63.5	66.8	70.1	**70.7**	**70.2**	**71.4**

The highest three results and the names of proposed methods are bold-faced.
Abbreviations: ESA: Expressway Service Area; ETS: Expressway Toll Station; GTF: Ground Track Field.

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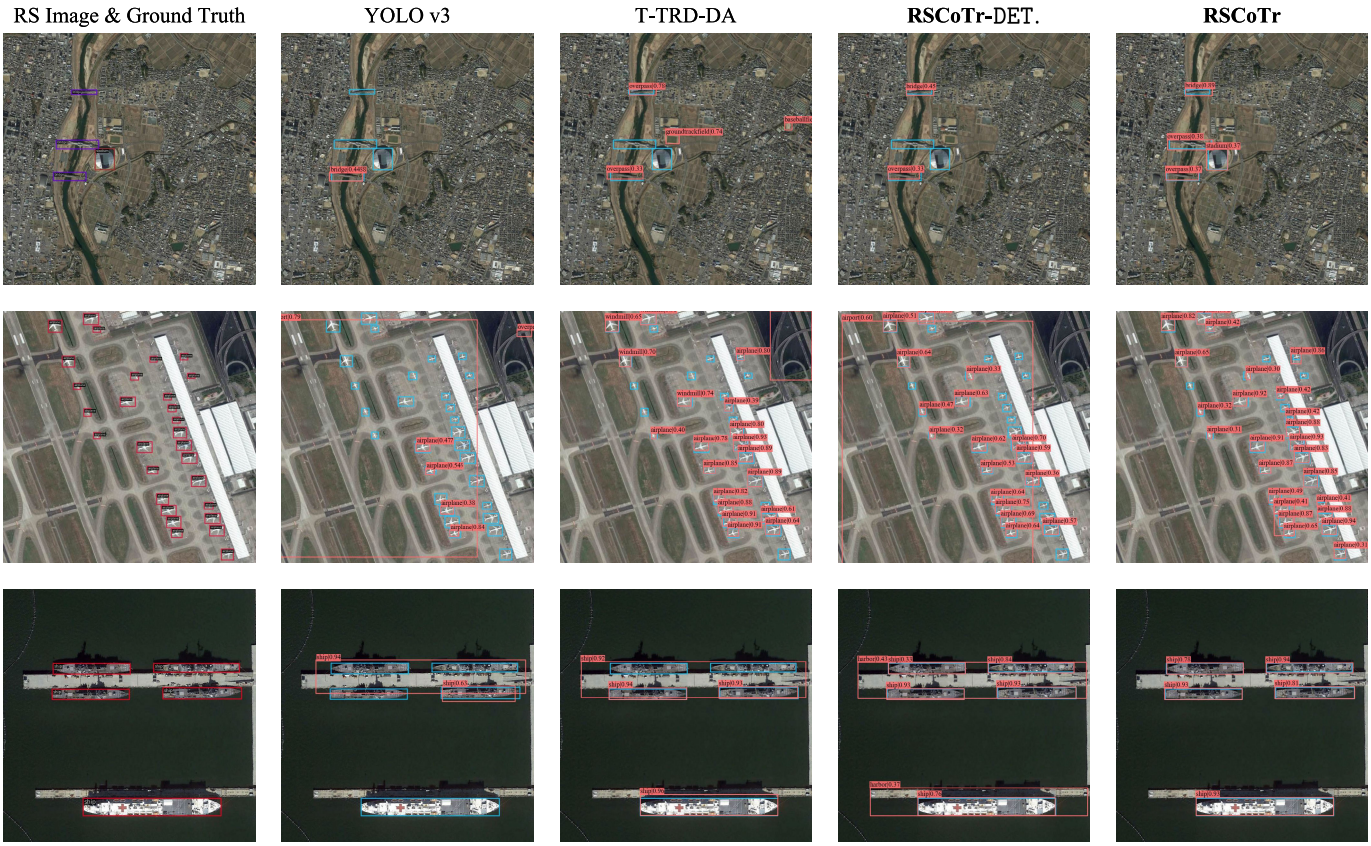


Fig. 13. Qualitative results of the multitask learned RSCoTr, the single-task learned RSCoTr-DET., a transformer-based detector (T-TRD-DA [13]), and a CNN-based detector (YOLO v3 [72]). For the first row, the ground-truth targets of different categories are enclosed with bounding boxes of different colors. For the other four rows, the ground-truth targets are all surrounded by blue boxes and the predictions are marked with red boxes. Besides, the classification scores of each targets are given, whose filter threshold is 0.3.

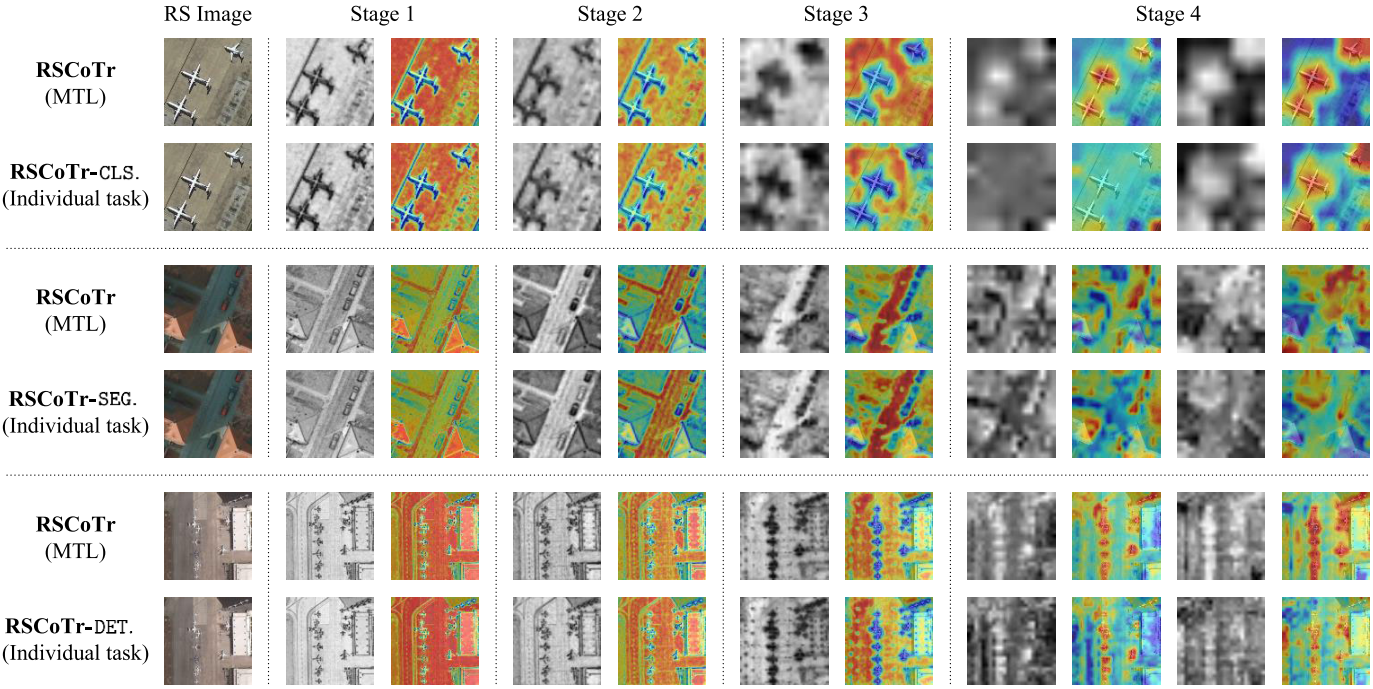


Fig. 14. Visualization of various feature maps, including the four stages features of the RSCoTr and individual task baselines on the three datasets of the corresponding tasks.

perform better than those with only a single scale. The proposed baseline and MTL methods both achieve highest per-classes APs. Besides, the RSCoTr achieves the best mAP result. Both the multitask learned RSCoTr and the single-task

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learned RSCoTr-DET. beat the transformer-based T-TRD-DA without using the strong data argumentation. The RSCoTr surpasses the state-of-the-art detectors, MSFGNet [75] and GLNet [76], by 1.3% and 0.7%, respectively. Moreover, the MTL brings to the RSCoTr more precise results on most categories and a significant improvement on mAP, which demonstrates that the MTL promotes the detection. Fig. 13 shows several comparison examples of the qualitative results, which illustrate that the multitask learned RSCoTr detects almost all targets and has less false alarm cases.

F. Visualization

To interpret the proposed method intuitively, we select one RS image from each dataset for visualization. The shared features of all stages are extracted and compared with the conventional features of single task learning, which are shown in Fig. 14. The feature maps in the even rows are individual features of the single-task learning, while those in the odd rows are the shared features of the MTL. And the three pairs of rows are the feature maps of the sample in the RESISC classification dataset, DIOR detection dataset, and Potsdam segmentation dataset, respectively. The figures in the first column are the input RS images. Note that the input RS image from the segmentation dataset is the split patch. In addition to the first column, the feature maps in the even columns are visualized directly where lighter indicates higher values and vice versa, and those in the odd columns are heat maps of resized features superimposed on the original RS image where warmer indicates higher values and vice versa. The four pairs of columns represent the four feature stages. For each stage, a random channel of the output feature maps is visualized. And for each image and task, the feature maps of the same channel are visualized for comparison. All the visualization maps are upsampled to the size of the input RS image with binary linear interpolation.

As is illustrated in Fig. 14, the features of the RSCoTr and the single task baselines are similar in shallow stages and discrepant in deep stages. It is difficult to visually distinguish the feature maps of the multitask model and each single-task baseline in the former two stages. The texture of the shallow feature maps is similar to that of the input RS images, hence, these features contain more high-resolution information, which is appropriate for spatially fine-grained predictions like semantic segmentation and tiny object detection. For the deeper stages, the feature maps become nonobjective. It is hard to observe the feature in the map visualized directly, because the texture becomes indistinct and the spatial resolutions are smaller. The heat maps are more helpful to distinguish which regions of the input RS image the network are more interested. The selected third stage of feature maps pays more attention to the scene background. The features of the fourth stage seem to be more focused on the targets like vehicles, airplanes, and buildings. It can be found by comparing the multitask learned RSCoTr and single-task baselines that the multitask model has better target localization capability. In the fourth stage of the first sample from the RESISC classification dataset, the ground-truth category is "airplain." The single-task baseline has small feature values in the first feature map. Besides, the baseline has biased localization of the right-top aircraft in the second feature map. In contrast, the multitask learned RSCoTr is more accurate for aircraft localization and also has high values. Hence, the learning of localization in detection and segmentation helps to recognize targets in the classification task. Besides, the deep feature values in regions of background of the samples from the other two sets and tasks are not as low as the single-task baselines, which mean that the classification task helps the dense prediction tasks to pay more attention to global scene and background.

V. Conclusion

In this study, the effectiveness of transformer for RS MTL was investigated. Specifically, a well-designed transformer and a proper co-training procedure were proposed for RS image classification, segmentation, and detection. Verification experiments were conducted on widely used datasets and several conclusions can be drawn from the experimental results:

1) The proposed MTL framework RSCoTr, which was composed of a shared transformer encoder and task-specific transformer decoder, could learn to process classification, segmentation, and detection task simultaneously. The RSCoTr had the advantages of less parameters, more uniform architecture, and better generalization capacity for different RS tasks.
2) To train an MTL model with datasets of each individual task, the co-training procedure was proposed as easy as training for a single RS task. The proposed co-training procedure was flexible so that various learning strategies could be exploited.
3) According to the experiments, with a well-designed learning strategy, the RSCoTr of co-training for multiple tasks achieved better performance than of individually training for a single task, which demonstrated the large potential of MTL for RS. Besides, the RSCoTr achieved state-of-the-art performance on all the learned tasks when comparing to the specialized methods of each task, which shows the effectiveness of the proposed methods.

Although it is an early attempt of the idea of transformer-based method for RS MTL, this study opens a new window for co-training transformer to handle multiple RS tasks.

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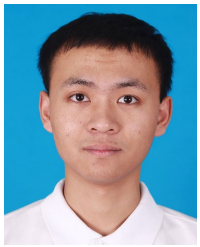
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